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# Remotely assessing foundational skills of 5–14-year-old children: A six-country psychometric evaluation of the remote assessment of learning (ReAL)

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## Abstract

**Approximately 250 million children worldwide are out of school. There is growing consensus for investing in feasible, contextually appropriate, psychometrically tested, remote tools to support quality education in low-resource contexts, including low- and middle-income countries and crisis-affected contexts. Save the Children developed the Remote Assessment of Learning (ReAL) to assess 5–14-year-old children's foundational learning. Children (*N* = 4,840*)* were sampled from Cambodia, Mozambique, Niger, the occupied Palestinian territories, the Philippines, and Sudan, with an approximately equal proportion of male and female children within each country. The study assessed inter-rater reliability, factor structure, item slope and difficulty, criterion validity, and test–retest reliability. Results show moderate evidence that ReAL is valid and reliable for literacy and numeracy; evidence for social-emotional skills is weaker. This is the first cross-country evaluation of a remote assessment of learning.**

**Keywords** foundational skills, remote assessment, child development

## Lay summary

Approximately 250 million children worldwide are not currently attending school. To help improve education in places with limited resources or those affected by crisis the Remote Assessment of Learning (ReAL) was created to assess children aged 5 to 14 on key skills such as reading, math, and social-emotional learning. This study tested ReAL with 4,840 children from six countries: Cambodia, Mozambique, Niger, the occupied Palestinian territories, the Philippines, and Sudan. The study found that the ReAL is

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a moderately reliable and valid measure of children's literacy and numeracy skills, though the results for social-emotional skills were less clear. This is the first study to evaluate a remote learning assessment across multiple countries, showing promise for using such tools to support education where in-person assessments are not possible.

# Introduction

Adopted by all United Nations Member States, Sustainable Development Goal (SDG)[1] 4 commits the global education community to ensuring that all children can achieve essential holistic learning and development outcomes through their education. This commitment can only be upheld through investments in the most educationally marginalized children: The approximately 251 million children worldwide who are out of school and the millions more not learning in school, an issue further exacerbated by conflict and crisis (UNESCO, 2024). The United Nations estimates that by 2030, 300 million students will lack the basic numeracy and literacy skills essential for full participation in today's world (United Nations, 2023). Global stakeholders increasingly recognize the need to invest in field-feasible, contextually appropriate, and psychometrically robust measurement tools to advance quality education in low-resource settings, including conflict-affected and low- and middle-income countries (LMICs; Tubbs Dolan, 2019). Such tools generate accurate and timely data—often described as the "lifeblood" of the SDGs (Sachs, 2012, p. 2210)—on key dimensions of children's learning and holistic development, enabling evidence-based decision-making within education systems.

In the last two decades, widespread use of orally administered assessments of learning for pre-primary and primary school-aged children in LMICs has helped policymakers and educators understand children's progress in reading, numeracy, and increasingly in social-emotional learning (SEL) (Montoya et al., 2016; Mulligan & Ayoub, 2023; Sowa et al., 2021). The development of robust assessments is an important scientific inquiry and a moral imperative to safeguard children's right to quality education. Such culturally relevant and scalable assessments are needed to better understand learning gaps of children—a need that is underscored by the high economic costs associated with a lack of formal education (UNESCO, 2023).

While there has been growth in the development and testing of learning assessments administered face-to-face, there is limited evidence for learning assessments administered remotely. There is an urgent need to develop and test such assessments, as educators, officials and humanitarian actors in low-resource and crisis-affected settings have consistently faced the challenge of how to assess children's academic outcomes and social-emotional skills when lack of resources, lockdowns, school closures, and other unexpected crises prevent the use of face-to-face assessment tools. The 2020 COVID-19 pandemic exacerbated and highlighted this challenge, as one analysis found that 214 million students from pre-primary to upper secondary education in 23 countries missed at least three quarters of classroom instruction time over one academic year due to school closures (UNICEF, 2021). Distance learning programs were implemented, but practitioners lacked a valid, reliable, relevant, and feasible remote assessment tool to evaluate students' growth of academic and social-emotional skills when engaging with these programs. There are several reasons why remote assessments are needed. Leave no one behind is a key transformative promise of the 2030 SDGs. To realize this promise, we need to find ways to measure foundational learning skills. Remote assessment is vital for marginalized (e.g., out-of-school children) specifically because they lack access to traditional assessment systems. Moreover, a recent UNESCO report (UNESCO, 2024) documents stark regional disparities in the rates of out-of-school children between low-income and high-income countries. In conflict and crisis settings, in particular, monitoring is often discontinued due to a lack of security and other urgent humanitarian priorities. Without recognizing what these children know and can do, education systems cannot provide adequate and targeted support. Therefore, these children remain largely invisible in the developmental literature, in educational planning, and in resource allocation. Developing and validating remote assessments offers a scalable pathway to generating actionable data on foundational learning skills for children that conventional, face-to-face assessment methods fail to reach. Without validated remote assessments, education systems lack the tools to identify learning gaps and evaluate distance learning programs and other interventions. Moreover, face-to-face assessment of children in low-resource contexts can be immensely expensive, which means that marginalized children in under-resourced systems are at risk of getting assessed last and least. Here, remote assessments are needed to provide immediate, actionable feedback to those closest to children (e.g., caregivers, community workers).

Most of the existing remote learning assessments have been created for the adult or elder learning space (Rapp et al., 2012 ; Sticht et al., 1996), or use a hybrid model that includes both in-person and remote assessment (Aker & Ksoll, 2020 ). To our knowledge, there are only two examples of fully remote, phone-based assessments that have been used to assess learning in LMICs: a numeracy assessment that was developed by Angrist et al. (2020) and tested in Botswana among children in grades 3–5 that was found to accurately capture basic numeracy skills, and a language and literacy assessment conducted by Sobers et al. (2023) in Cote d'Ivoire among children in grade 5 that was found to be valid and reliable. In response to this gap in the literature and programmatic need, Save the Children developed the Remote Assessment of Learning (ReAL) tool to remotely assess literacy, numeracy, and social-emotional foundational skills for children ages 5–14-years-old.

The development of the literacy and numeracy domains for the Remote Assessment of Learning (ReAL) drew heavily on the Early Grade Reading Assessment (EGRA) and Early Grade Mathematics Assessment (EGMA) frameworks, which provided validated, psychometrically sound constructs for assessing foundational learning skills. These tools, developed by RTI International with support from the United States Agency for International Development, have been widely implemented across more than 100 countries, offering a robust evidence base for their reliability and cross-cultural applicability (Gove & Wetterberg, 2011; RTI

[1] The Sustainable Development Goals (SDGs) are a set of 17 global objectives adopted by the United Nations to achieve a better and more sustainable future for all, addressing challenges such as poverty, inequality, climate change, environmental degradation, peace, and justice by 2030.

International, 2014, RTI International, 2015). In the ReAL study, EGRA and EGMA constructs served as the foundation for item development. Using these constructs ensures conceptual comparability across countries, while maintaining flexibility for contextualization through adaptation of language, cultural references, and examples to local curricula and everyday experiences. This balance between global standardization and local relevance enhances the validity of cross-country comparisons and the interpretability of results within each context, supporting both global learning monitoring and national education system strengthening efforts (Dubeck & Gove, 2015; Platas et al., 2014).

The development of the SEL items for ReAL was based on the core competencies model of social-emotional learning proposed by the Collaborative for Academic, Social, and Emotional Learning (CASEL) and a body of literature focused on understanding child development within unique cultural contexts (e.g., Eisenberg et al., 2001; Oburu et al., 2016 ; Panter-Brick et al., 2018 ; Yoshikawa & Way, 2008). ReAL's SEL items also align with three- and five-dimensional taxonomies of social-emotional skills and competencies other than CASEL (see Soto, Napolitano, & Roberts, 2021, for a concise overview). One key aim was to infuse global policy, developmental, and measurement research with culturally appropriate developmental science insights, ensuring that proposed measures were psychometrically robust, developmentally driven, and culturally sensitive (e.g., Barbot et al., 2020; Halpin et al., 2019; Jordans et al., 2013; Wuermli et al., 2015; Yoshikawa et al., 2015; Yousafzai et al., 2014).

The ReAL Network, comprised of Save the Children, local academic partners, and a global consortium of thematic practitioners and researchers with psychometric expertise, developed the ReAL cross-cultural validation study, which targeted hard-to-reach children (e.g., children impacted by school closures, children not enrolled in school) in seven different LMICs. The validation approach was designed to address three root causes: (1) a lack of rigorous evidence on remotely administered assessments in LMICs (Angrist et al., 2020 ); (2) a gap in research-to-practice mechanisms in LMICs due to limited resources to translate research into policy or practice (Shumba & Lusambili, 2021); and (3) many of the measures used in LMICs originate from educational research in high-income countries (e.g., the EGRA), and their validity and reliability in crisis contexts remain uncertain (Bartlett, et al., 2015; Dowd et al., 2019; Halpin & Torrente, 2014).

ReAL is built upon assessments such as the Holistic Assessment of Learning and Development Outcomes (Krupar et al., 2019; Krupar & D'Sa, 2024), the International Development and Early Learning Assessment (IDELA; Halpin et al., 2019; Pisani, et al., 2018; Wolf et al., 2017), the International Social-Emotional Learning Assessment (D'Sa et al., 2019), EGRA, and EGMA to ensure that tests were previously validated in face-to-face settings and that the assessment structure is familiar to implementers who are often familiar with the mentioned commonly used assessments. The thorough and systematic development of ReAL was, thus, based on assessments with well-documented psychometric properties across samples from diverse countries and contexts. Yet, cultural differences play an important role in the contextualization of ReAL. In the present study, standardized administration guidance and design workshops with local academic partners informed the contextualized item adaptation process. Items and stimuli were adapted based on locally familiar concepts, national curricula, and grade-level textbooks appropriate to a given subdomain. The translation of items to local language prioritized translation of the item's essence using simple language and similar content rather than literal translation. The importance of cultural differences is particularly pertinent when the goal is to establish the cross-country comparability of items measuring social and emotional skills. Recent studies have documented the possibility of developing SEL assessments that meet measurement invariance requirements when compared across countries (e.g., a self-assessment measuring social skills of youth in Uganda and Guatemala; Menezes Cunha et al., 2021; the comparability of the measurement model underlying the 2018 OECD Survey on Social and Emotional Skills across two countries in East Asia and two countries in North America; Lee & Junus, 2024). For the ReAL study, SEL stimuli were adapted based on locally accepted social concepts, and local partners developed a master list of appropriate and inappropriate responses linked to relevant items that assessors used to code child responses. Adaptation of items maintained the format and the content of the original questions.

ReAL also addressed several crucial challenges of measuring social and emotional skills, for instance, by using culturally grounded, hypothetical, yet naturalistic vignettes for oral administration instead of relying on Likert scales. This approach has been recommended to account for the potential impact of different cultural backgrounds on item responses (e.g., Abrahams et al., 2019). The approach to involve practitioners and their priorities and needs in the contextualization of ReAL and the design and implementation of the present validation study has been pointed out as a major opportunity to advance SEL measurement (e.g., McKnown, 2019). The wide 5–14-year age range was used at the pilot stage, as many of the instruments from which ReAL items are drawn had been used with children of these ages, and in many of the contexts in which this instrument would be used, there are children in this age range learning foundational skills. Table 1 outlines the subdomains for the Remote Assessment for Learning as well as the number of items for each subdomain and a description of the subdomain.

After the first iteration of the tool was developed, ReAL was piloted in Bangladesh, Guatemala, the Philippines, and Zambia. This first (alpha) pilot found high internal consistency reliability within the literacy (minimum Cronbach's $\alpha = .86$), numeracy (minimum Cronbach's $\alpha = .78$), and social-emotional learning (minimum Cronbach's $\alpha = .90$) domains. This indicated that the tests reliably assess the hypothesized domains and sub-domains in pilot sites. The alpha pilot identified some sub-domains that did not function well psychometrically or were difficult to administer based on feedback from pilot sites. These sub-domains and items were either removed or revised. Given the initial evidence that the ReAL has the potential to be a reliable and valid measure of these learning skills, the team embarked on the beta phase using the revised tool, which involved a rigorous validation study in seven countries with hard-to-reach populations: Cambodia, El Salvador, Mozambique, Niger, the Occupied Palestinian Territory (oPt), the Philippines, and Sudan. All countries, except El Salvador, selected the High Access version of ReAL (see Measures section below, for more detail), which is the focus of this study.

**Table 1** ReAL domain and subdomain description.

| **Literacy**<br>**Age Range: 5–14 (administered all items)**<br>**Translations: French, Spanish, Arabic, Khmer, Portuguese, Tagalog** | | | |
|---|---|---|---|
| **Oral Language Module** | | | |
| **Subtask** | **Subdomain Description** | **No. of Items** | **Source of Items** |
| Expressive Vocabulary | Name common foods/animals | 2 (up to 20 responses recorded) | IDELA |
| Listening Comprehension | Number of correctly answered questions following a short story | 5 | IDELA |
| Retelling a Story | Level of syntactic complexity of story retelling | 2 | USAID ELM (USAID, n.d.) |
| Reading Module | | | |
| Letter/letter sound identification | Number of correctly identified letters/letter sounds | 20 | EGRA |
| Common Word Identification | Number of single words read correctly | 10 | EGRA |
| Sentence-level Comprehension | Number of read phrases correctly marked as true or false | 5 | EGRA |
| Oral Passage Reading | 1. Ability to read independently: Ability to read at least 5 words correctly in 30 s<br>2. Fluency: Number of words in a short story read correctly in a minute<br>3. Accuracy: Percentage of words in a short story read correctly<br>4. Comprehension: Questions related to a short story read aloud by student or assessor | ~100 words<br>7–10 comprehension questions | EGRA |
| **Numeracy**<br>**Age Range: 5–14 (administered all items)**<br>**Translations: French, Spanish, Arabic, Khmer, Portuguese, Tagalog** | | | |
| **Subtask** | **Subdomain Description** | **No. of Items** | **Source of Items** |
| One-to-one Correspondence | Ability to identify the correct number of items in a set | 3 | EGMA |
| Number Identification | Number of correctly identified numbers | 12 | EGMA |
| Addition | Ability to correctly add one- and two-digit numbers | 10 | EGMA |
| Subtraction | Ability to correctly subtract one- and two-digit numbers | 10 | EGMA |
| Word problems | Ability to correctly answer numerical word problems | 3 | EGMA |
| **Social and Emotional Learning**<br>**Age Range: 5–14 (administered all items)**<br>**Translations: French, Spanish, Arabic, Khmer, Portuguese, Tagalog** | | | |
| **Self-Concept** | | | |
| **Subtask** | **Subdomain Description** | **No. of Items** | **Source of Items** |
| Self-Concept | Positive hope<br>Realistic understanding of obstacles<br>Realistic understanding of supports | 6 | HALDO |
| Relationships | | | |
| Use of Social Supports | Identification of peers and adults to support different experiences | 4 | ISELA |
| Perseverance | | | |
| Help Seeking Behavior | Talking to others when working on something difficult | 4 | ISELA |
| Stress Management | | | |
| Social Supports | Talking to others when sad | 4 | ISELA |
| Behavioral Regulation | Self-soothing practices | 3 | ISELA |
| Empathy | | | |
| Identifying Feelings of Others | Awareness of others' emotions<br>Feeling recognition | 5 | ISELA |

(*Continued*)

Table 1 (Continued)

**Social and Emotional Learning**
**Age Range: 5–14 (administered all items)**
**Translations: French, Spanish, Arabic, Khmer, Portuguese, Tagalog**

**Self-Concept**

| Subtask | Subdomain Description | No. of Items | Source of Items |
|---|---|---|---|
| Cognitive Empathy | Perspective taking<br>Awareness of others' expectations | 6 | ISELA |
| Conflict Resolution | | | |
| Social Problem Solving | Strategies and methods to negotiate interpersonal disputes with peers | 4 | ISELA |
| Interpreting Hostility | Identification of non-hostile response during conflict | 2 | ISELA |

*Note.* IDELA = International Development and Early Learning Assessment; ISELA = International Social and Emotional Learning Assessment.

# The present study

This present study aims to examine the psychometric properties of the High Access modality of ReAL.[2] High access refers to situations where children and caregivers have access to advanced technology, such as a smartphone capable of using WhatsApp or an audio phone paired with assessment-specific learning materials (hard copy). In these cases, assessments are conducted directly with the child via telephone. Following contextualization and training, assessors verbally guided the caregiver and child through a detailed protocol over the phone while assessment stimuli were shared via the phone using text messaging, WhatsApp, or a hard copy handed out to the caregivers prior to the assessment.

The study's specific objectives are, across the six-country sample, to (1) assess the inter-rater reliability of ReAL; (2) identify the underlying factor structure of ReAL and assess the practical relevance of the tool by understanding each item's characteristics; (3) evaluate the test–retest reliability of ReAL; and (4) understand the criterion validity by measuring the extent to which the ReAL domains correlate with existing non-remote assessments of learning.

# Method

## Participants

Participants were sampled from Cambodia ($n = 1{,}108$), Mozambique ($n = 458$), Niger ($n = 854$), oPt ($n = 1{,}135$), the Philippines ($n = 798$), and Sudan ($n = 587$) (Figure S1.1). There was an approximate 50/50 split in child sex within each country. However, slightly more males were sampled in Mozambique (53%), while more females were sampled in Sudan (65%), the Philippines (54%), Cambodia (52%), Niger (53%), and oPt (51%). The sex of the caregiver differed by country, with most caregivers in oPt, the Philippines, Cambodia, and Niger being female, while most caregivers in Mozambique were male. In Sudan, approximately 50% of the sampled caregivers were male and 50% were female. Mothers primarily helped to facilitate the application in oPt and the Philippines. In Mozambique mostly fathers helped to facilitate, while in Cambodia a variety of different caregivers helped (i.e., 47% mothers, 28% fathers, 10% grandmothers). The average age of the child ranged by country (Cambodia = 9.15 years with *SD* of 2.25 years; Mozambique = 13.23 years with *SD* of 2.06 years; Niger = 10.75 years with *SD* of 2.13 years; oPt = 10.04 years with *SD* of 2.38 years; Philippines = 12.68 years with *SD* of 1.80 years; Sudan = 11.44 years with *SD* of 2.62 years). The minimum age of a child sampled was 4.30 years in Sudan, and the maximum age was 21.32 years in Mozambique. We included children older than 14 years of age who attended the same grade level as the selected sample but whose formal schooling was interrupted and, as a result, who re-entered school at an older age than the nominal age for the respective grade level.

[2] The ReAL High Access tool is freely available for sharing and adapting under a Creative Commons CC BY-NC-SA 4.0 license.

# Procedure

## Sampling

The target sample was 200 caregiver-child dyads each within five age groups: 5–6-year-olds, 7–8-year-olds, 9–10-year-olds, 11–12-year-olds, and 13–14-year-olds. The sample size was determined based on the statistical power required to conduct confirmatory factor analysis (CFA). Three considerations were applied to estimate the target sample size: (1) the parameter estimates from the one-factor CFA of subtests in the literacy, numeracy, and SEL domain, using caregiver-reported data from Bangladesh ($n = 323$ observations of 5–12-year-old children); (2) expected non-normal item score dispersion (i.e., ceiling effects); (3) the binary or ordinal scale of measurement of the manifest indicators; (4) feasibility to collect data from hard-to-reach (e.g., children not enrolled in school or education programs) in six LMICs. The target sample size of 1,000 children (if sampled across the entire age span) is comparable to published research using similar measures such as IDELA (e.g., Halpin et al., 2019; Pisani et al., 2018; Shavitt et al., 2022; Wolf et al., 2017). Each country team decided what modality to test based on access to smartphones, assessment of feasibility, acceptability, intended use, and, in some cases, policy priorities (e.g., ministry partners interested in a particular application). Cambodia, Mozambique, Niger, oPt, the Philippines, and Sudan all opted for the high-access modality.

### Country considerations in sampling

Across all samples, inter-rater reliability, test–retest reliability, and criterion samples were randomly selected in each country, and each represented 20% of the overall sample. Children who were randomly assigned to the test–retest sample were visited

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again between 1–12 weeks after their initial assessment. This timeframe varied by country and was deemed (1) feasible for the country teams to re-contact and re-assess families in areas with restricted access, (2) appropriate to analyze whether ReAL produces consistent results and stable measures over a short period (i.e., rank-order stability of children's test scores over time), which is important for evaluating the impact of educational programs in these settings, and (3) comparable to the analysis of retest reliability of existing assessments of foundational learning skills (e.g., Outhwaite, Aunio, Leung, & Van Herwegen, 2024; Pisani, Borisova, & Dowd, 2018). Table S.0 contains additional information about the sampling approach in each country.

Children who were randomly assigned to the inter-rater reliability sample were assessed by two data collectors who independently recorded their scoring of the child's responses. Data collectors (i.e., enumerators) were trained in a multi-day training workshop to familiarize themselves with the child safeguarding protocol, tool modality, interviewing techniques, scoring guidelines, and data collection using a form programmed in KoBo Toolbox (n.d.). Moreover, the data collectors were trained on how to mitigate the potential influence of the caregiver on the assessment process and the child's response. ReAL is designed to allow for some degree of caregiver scaffolding. This is particularly important for children who are not familiar with a standardized assessment process. However, caregiver influence (e.g., prompting a correct response) was not reported by the data collectors. We obtained ethical approval through Save the Children's U.S.-based IRB because the ReAL validation was a single, sponsor-led, multi-country study implemented under a harmonized protocol (standardized instruments, consent/assent scripts, enumerator training, safeguarding and adverse-event/referral procedures, and data security). A centralized review ensured a consistent ethical "floor" for protecting children and caregivers across all six countries. It also aligned oversight with the institution accountable for study governance, safeguarding standards, and cross-border data management. Pursuing full IRB review separately in each country was not feasible or proportionate because local ethics infrastructure and requirements varied substantially (some countries have established IRBs, others do not or have limited access for minimal-risk social/behavioral research). A country-by-country approach could have produced uneven standards and delays unrelated to participant safety. The study was minimal risk (non-invasive assessment and interview procedures comparable to routine educational activities), making centralized review an appropriate and efficient mechanism to ensure rigorous protections. Reliance on Save the Children's IRB did not replace local authorization or contextual safeguards. In each country, we sought and documented appropriate local permissions (e.g., Ministry/district/school approvals and partner-institution clearances) and adapted materials for language and context without changing core protections.

The specific sampling strategy, translation, and adaptation process differed by country. ReAL was carefully adapted for the context through a collaborative process led by the local academic partner and the Save the Children office with guidance from the Global Team. Tool adaptation followed the guidance in the administrative guide and involved translating the tool into the official language of instruction and contextualizing its content, stories, and stimuli to align with the cultural and linguistic realities of the country while maintaining its core measures. The global team supported this process by sharing insights from the alpha phase, reviewing suggested contextual adaptations and guiding decisions on sampling and contextualization approaches.

## Measures

### The remote assessment of learning (ReAL)

ReAL is a remote assessment of learning intended to measure children's literacy, numeracy, and SEL skills for monitoring and evaluation purposes among 5–14-year-olds. The literacy domain contains an oral language module (with expressive vocabulary, listening comprehension, and retelling a story sub-domains) and a reading module (with letter/letter sound identification, common word identification, sentence-level comprehension, and oral passage reading sub-domains). Each sub-domain contains between 2–10 questions that are typically scored dichotomously as correct or incorrect, except for expressive vocabulary which is a continuous answer. The numeracy domain contains five sub-domains (one-to-one correspondence, number identification, addition, subtraction, and word problems). Each sub-domain contains between 3–10 questions, and all answers are scored dichotomously as correct or incorrect. The SEL domain contains six sub-domains (self-concept, relationships, perseverance, stress management, empathy, and conflict resolution). Each sub-domain has between 4–11 questions, and each answer is scored dichotomously as correct or incorrect, or as appropriate or inappropriate.

ReAL can be administered over the phone in three different administration modalities (high access, low access, or caregiver-reported) depending on the extent of child accessibility, materials (stimuli) availability, and type of phone (smart or conventional) that the parents/caregivers have access to. While high access is appropriate to situations where households can access technology such as smartphones, or audio phones paired with official hard copy assessment stimuli, low access provides an assessment modality where households can access audio phones and use existing materials at home for the assessment stimuli. The caregiver-reported modality is appropriate where caregivers have audio phone access and children are not available for direct assessment. These modalities ensure that ReAL is accessible to everyone, regardless of their technological resources (Save the Children, n.d.a, n.d.b).

### Criterion measures

The criterion measures varied by country. In Mozambique, Niger, oPt, and the Philippines, the EGRA (RTI International, 2015) and the EGMA (RTI International, 2014) were used for measuring literacy and numeracy. These measures were selected as criterion measures, as they are widely used across LMICs, including in Mozambique, Niger, oPt, and the Philippines. They also have been used with children across the primary school age range as measures of foundational reading and math skills. For SEL, an adapted version of the Psychosocial Support Scale, previously validated in Sudan (Olayemi et al., 2021), was used as a criterion measure in Mozambique, Niger, oPt, and the Philippines. This measure most closely matched the constructs being measured by ReAL in the absence of country-specific measures. In Cambodia, the East Asia-Pacific Early Childhood Development Scales (EAP-ECDS) 2018 was used as a criterion measure for all constructs (Rao et al., 2024).

## Statistical analysis

Analyses were conducted in Stata v18 (Stata- Corp, College Station, TX, USA) and R (R Core Team, 2021). Descriptively, we analyzed the age distribution, grade distribution, languages spoken at home, relationship of caregiver to child, and caregiver sex by country. Spearman correlation coefficients were then calculated between each ReAL sub-domain.

Inter-rater reliability between two independent enumerators was calculated using the Krippendorff's α (Marzi et al., 2024) to account for various combinations of enumerators. An α of 1 indicates perfect agreement, .80–.99 indicates a satisfactory level of agreement indicating a reliable rating, .67–.79 indicates a lower bound for tentative conclusions and < .67 indicates poor agreement among raters (Marzi et al., 2024). Spearman correlations were computed at the sub-domain level for test–retest reliability. For test–retest correlations, we used Cicchetti's (1994) recommendations for appraising the magnitude of correlations ranging between .40–.59 as fair, .60–.74 as good and above .75 as excellent. For both the inter-rater reliability and test–retest analyses, missing values among individuals with partial missingness were assumed to be incorrect. Individuals missing all information on a given sub-domain were dropped from that sub-domain analysis but included in other analyses where they had at least partial missingness. Missing data ranged from <1% to 30% depending on the question.

We used the lavaan package (Version 0.6-17; Rosseel, 2012) embedded in the R environment to perform CFA for each sub-domain based on the a priori, theoretical mapping of the ReAL items onto the literacy, numeracy, and SEL domains. Given the a priori assumption that the factor structure is not qualitatively different across countries, a pooled structural equation model was run across countries. Specifically, a one-factor CFA model was conducted separately for listening comprehension (5 items), letter/letter sound identification (20 items), common word identification (10 items), sentence-level comprehension (5 items), oral passage reading comprehension (7 items), number identification (12 items), addition (10 items), subtraction (10 items), self-concept (6 items), use of social supports (4 items), and help-seeking behavior (4 items). A two-factor CFA model was developed separately for stress management (four items loaded on social supports and three different items loaded on behavioral regulation), empathy (five items loaded on identifying feelings of others and six items loaded on empathic responding), and conflict resolution (four items loaded on social problem solving and two items loaded on interpreting hostility). Separate CFAs were conducted to appraise the unidimensionality of the ReAL subtests. More complex models such as higher-order or bi-factor models involving item-level data from multiple subtests across multiple domains were not estimated because of the limitations of statistical power. CFA was not conducted for expressive vocabulary (two items), retelling a story (two items), one-to-one correspondence (three items), and word problems (three items) since a one-factor CFA model with two indicators was under-identified and a one-factor CFA model with three indicators was just-identified. We used the weighted least square mean and variance estimator to generate accurate inferences for binary indicators. Pairwise deletion was used to deal with missing data. Global goodness of fit was evaluated based on chi-square ($\chi^2$), Comparative Fit Index (CFI), Root Mean Square Error of Approximation (RMSEA) and its 90% confidence interval, and Standardized Root Mean Squared Residual (SRMR). Given that conventional CFI and RMSEA tend to be overestimated when indicators are ordered-categorical variables, we reported robust CFI and RMSEA proposed by Savalei (2021) when available. Values greater than .90 and .95 for both the CFI and TLI, respectively, are considered to reflect acceptable and excellent fit to the data, whereas values for RMSEA less than .05 and .08 indicate excellent fit and acceptable fit to the data, respectively (Hu & Bentler, 1999; Marsh et al., 2005). An SRMR value of less than.08 was recommended (Hu & Bentler, 1999). Moreover, local model fit was evaluated by examining Bentler-type correlation residuals (type = "cor.bentler" in lavaan), which were reported in Tables S1.2–S1.12 for each CFA model.

For sub-domains of literacy and numeracy with empirical evidence supporting unidimensionality, we proceeded with appraising item characteristics by conducting item response theory (IRT) analysis. IRT was not conducted for SEL because the item parameters for SEL are not as informative as those for literacy and numeracy. Unlike literacy and numeracy that are often more clearly defined and objective, SEL constructs can be more subjective and complex, involving attitudes, behaviors, and emotions. These constructs might not fit well into the IRT framework, which assumes unidimensionality. Pragmatically, knowing the slope and the difficulty for each SEL item, for example, "Can you describe what you hope will happen in your life in the future" (item sel 1), does not provide valuable information for refining the item. Two Parameter Logistic (2PL) IRT model characterizing item slope and difficulty was carried out using the *mirt* package (Version 1.41; Chalmers, 2012) embedded in the R environment. Due to the relatively large number of sub-domains under examination and the number of items involved, we presented the item information function (IIF) and test information function (TIF) instead of presenting parameter estimates. This is because IIF and TIF offer straightforward ways to visualize item- and test-level performance.

Criterion validity was assessed by calculating Spearman correlation coefficients between the ReAL domains with their corresponding criterion measures for relevant literacy, numeracy, and social-emotional learning sub-domains.

# Results

## Correlations

ReAL sub-domains were all highly correlated with each other (Table 2). The strongest positive correlation was observed between subtraction and addition (Spearman correlation coefficient [$r_s$]= .680) and help-seeking behavior and use of social supports ($r_s$ = .661). The weakest correlation was seen between stress management, social support, and reading comprehension ($r_s$ = .146).

## Inter-rater reliability

Inter-rater reliability across domains was moderate to high, apart from the SEL domain in Sudan and Mozambique (Table 3). Krippendorff's α values in the literacy domain ranged from moderate agreement in Sudan (α = .781) to satisfactory agreement in all other countries. Krippendorff's α values in the numeracy domain ranged from moderate agreement in Mozambique

**Table 2** Pooled spearman correlations between ReAL sub-domains ($N = 4,840$).

| | Expressive vocabulary | Listening comprehension | Retelling a Story | Letter Identification | Common word Identification | Sentence Comprehension | Reading Comprehension | One to one Correspondence | Number Identification | Addition | Subtraction | Word problems | Self-Concept | Use of social supports | Help seeking behavior | Stress management, social supports | Stress management, behavioral regulation | Identifying feelings of others | Empathy |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Expressive Vocabulary | 1.000 | | | | | | | | | | | | | | | | | | |
| Listening Comprehension | 0.508 | 1.000 | | | | | | | | | | | | | | | | | |
| Retelling a Story | 0.458 | 0.560 | 1.000 | | | | | | | | | | | | | | | | |
| Letter Identification | 0.394 | 0.419 | 0.352 | 1.000 | | | | | | | | | | | | | | | |
| Common Word Identification | 0.491 | 0.517 | 0.497 | 0.571 | 1.000 | | | | | | | | | | | | | | |
| Sentence Comprehension | 0.406 | 0.466 | 0.423 | 0.494 | 0.599 | 1.000 | | | | | | | | | | | | | |
| Reading Comprehension | 0.486 | 0.417 | 0.449 | 0.444 | 0.636 | 0.489 | 1.000 | | | | | | | | | | | | |
| One to One Correspondence | 0.291 | 0.254 | 0.236 | 0.319 | 0.299 | 0.318 | 0.247 | 1.000 | | | | | | | | | | | |
| Number Identification | 0.315 | 0.298 | 0.261 | 0.534 | 0.494 | 0.456 | 0.426 | 0.365 | 1.000 | | | | | | | | | | |
| Addition | 0.330 | 0.368 | 0.361 | 0.479 | 0.551 | 0.511 | 0.459 | 0.314 | 0.535 | 1.000 | | | | | | | | | |
| Subtraction | 0.337 | 0.397 | 0.392 | 0.439 | 0.573 | 0.479 | 0.466 | 0.289 | 0.482 | 0.680 | 1.000 | | | | | | | | |
| Word Problems | 0.440 | 0.511 | 0.453 | 0.481 | 0.570 | 0.526 | 0.484 | 0.328 | 0.488 | 0.589 | 0.600 | 1.000 | | | | | | | |
| Self-Concept | 0.206 | 0.263 | 0.335 | 0.248 | 0.300 | 0.344 | 0.245 | 0.190 | 0.224 | 0.314 | 0.318 | 0.311 | 1.000 | | | | | | |
| Use of Social Supports | 0.245 | 0.260 | 0.207 | 0.248 | 0.223 | 0.242 | 0.164 | 0.205 | 0.239 | 0.242 | 0.235 | 0.302 | 0.201 | 1.000 | | | | | |
| Help Seeking Behavior | 0.290 | 0.290 | 0.250 | 0.286 | 0.272 | 0.295 | 0.226 | 0.225 | 0.284 | 0.290 | 0.281 | 0.346 | 0.229 | 0.661 | 1.000 | | | | |
| Stress Management, Social Supports | 0.244 | 0.267 | 0.183 | 0.238 | 0.212 | 0.199 | so | 0.192 | 0.224 | 0.213 | 0.224 | 0.289 | 0.172 | 0.628 | 0.558 | 1.000 | | | |
| Stress Management, Behavioral Regulation | 0.320 | 0.293 | 0.272 | 0.258 | 0.317 | 0.301 | 0.380 | 0.215 | 0.283 | 0.317 | 0.306 | 0.363 | 0.302 | 0.284 | 0.292 | 0.293 | 1.000 | | |
| Identifying Feelings of Others | 0.332 | 0.341 | 0.301 | 0.324 | 0.324 | 0.329 | 0.273 | 0.329 | 0.302 | 0.321 | 0.323 | 0.402 | 0.277 | 0.558 | 0.558 | 0.557 | 0.384 | 1.000 | |
| Empathy | 0.421 | 0.443 | 0.391 | 0.378 | 0.431 | 0.403 | 0.406 | 0.234 | 0.370 | 0.393 | 0.402 | 0.490 | 0.339 | 0.372 | 0.388 | 0.345 | 0.493 | 0.578 | 1.000 |

**Table 3** ReAL inter-rater reliability by country and domain.

| ReAL domain | Krippendorf's α | | | | | |
|---|---|---|---|---|---|---|
| | oPt ($n=185$) | Mozambique ($n=79$) | Philippines ($n=92$) | Cambodia ($n=208$) | Sudan ($n=355$) | Niger ($n=219$) |
| Literacy | .842 | .831 | .893 | .944 | .781 | .969 |
| Numeracy | .913 | .739 | .783 | .968 | .852 | .948 |
| Social emotional learning | .881 | .630 | .790 | .893 | .606 | .957 |

(α = .739) and the Philippines (α = .783) to satisfactory agreement in all other countries. Lastly, Krippendorff's α values in the SEL domain ranged from poor agreement in Sudan (α = .606) and Mozambique (α = .630) to moderate agreement in the Philippines (α = .790), to satisfactory agreement in all other countries.

## Factor structure

Table 4 presents CFA fit statistics for each sub-domain of literacy, numeracy, and SEL. The one-factor CFA model exhibited adequate global model fit indices in terms of CFI, RMSEA, and SRMR for listening comprehension and number identification. For common word identification, addition, and subtraction, the one-factor CFA model presented CFI and SRMR exceeding the recommended values, although RMSEA and its confidence interval failed to meet the recommended values. Although CFI and RMSEA did not exceed the recommended values for letter/letter sound identification and oral passage reading, SRMR did exceed the recommended values for oral passage reading. The one-factor CFA model for sentence-level comprehension presented inadequate fit statistics. The *p* values of the $\chi^2$ for use of social supports and help-seeking behavior indicated an excellent fit. Although the CFI for the two-factor CFA model for empathy exceeded the recommended value, it may be overestimated. This is supported by its relatively large SRMR. The proposed CFA models for self-concept, stress management, and conflict resolution produced negative residual variance, indicating that the estimates such as factor loadings and residual correlations are not reliable.

Factor loadings are presented in Supplementary Tables S1.2–S1.12. Except for oral passage reading, use of social supports, help-seeking behavior, and identifying feelings of others, standardized factor loadings were greater than .80 for all items. The standardized factor loadings for oral passage reading, use of social supports, and help-seeking behavior were all greater than .60 for all items. Standardized factor loadings for identifying the feelings of others ranged from .475 to.899.

Tables S2.1–S2.11 show the residual correlation matrix for each CFA model. For listening comprehension, letter/letter sound identification, common word identification, oral passage reading, number identification, addition, and subtraction, residual correlation coefficients were overwhelmingly close to or less than .10, suggesting that one latent factor sufficiently accounted for the covariances between indicators. For sentence-level comprehension, 4 out of 10 residual correlation coefficients were greater than .20, indicating one latent factor was not sufficient to account for the covariances between the five indicators. For the use of social supports, the residual correlation coefficient between two items (rel3 and rel4, as labeled in the open-access toolkit) was .480. For help-seeking behavior, the residual correlation coefficient between rel5 and rel6 was .449. The high residual correlation coefficient for the two sub-domains was caused by the strong association between rel3 and rel4 (tetrachoric correlation coefficient [$r_t$]= .99) and between rel5 and rel6 ($r_t$ = .99). For the two-factor empathy model, 13 out of 55 residual correlation coefficients were greater than.2.

Collectively, global model fit indices, standardized factor loadings, and local model fit indicate the existence of one latent factor for the literacy and numeracy sub-domains except for sentence-level comprehension. Despite some redundant items, the one-factor model for use of social supports and help-seeking behavior was supported. However, the constructs including self-concept, stress management, empathy, and conflict resolution were not empirically identified.

## Item and test characteristics

IIF and TIF for each sub-domain are presented in Figures S1.1-S1.7 and S2.1-S2.7, respectively. Despite the variability of items in distinguishing between children with different skills, the peaks for all IIFs were located at least one standard deviation below the mean. This suggests that these sub-domains were easy for the participants. Consequently, each sub-domain only provides reliable information for children whose skills are lower than average. This is evidenced by the TIFs. As a result, all sub-domains of literacy and numeracy may not be able to accurately assess skills for most children.

## Criterion validity

Criterion correlations were calculated for each country by sub-domain (Table 5, Table 6 and Table 7). Criterion correlations varied by country and sub-domain but were overall the strongest in the numeracy sub-domains across countries, and between countries were overall the strongest in Cambodia.

### Literacy

For the literacy domain, criterion correlations for letter/letter sound identification were satisfactory for Cambodia, with strong positive and significant correlations with the identifying consonants ($r_s$ = .702), identifying vowels ($r_s$ = .694), and consonant subscripts ($r_s$ = .720) sub-domains of the criterion tool. A similar positive correlation was found in Mozambique ($r_s$ = .368) and Niger ($r_s$ = .442). Letter/letter sound identification was not correlated with the corresponding criterion measure in oPt or the Philippines.

Common word identification was strongly and positively correlated with the reading words ($r_s$ = .782) and reading sentences ($r_s$ = .667) sub-domains in Cambodia. A positive correlation was

**Table 4** Pooled CFA fit statistics for each sub-domain of literacy, numeracy, and social and emotional learning .

| Sub-domain | $\chi^2$ (*df*) | *p* | *RMSEA*[90%*CI*] | *CFI* | *SRMR* |
|---|---|---|---|---|---|
| Literacy | | | | | |
| Listening comprehension | 5.966 (5) | .310 | .033 [.000, .090] | .999 | .007 |
| Letter/letter sound identification | 652.951 (170) | < .001 | .155 [.143, .167] | .886 | .033 |
| Common word identification | 309.506 (35) | < .001 | .158 [.141, .176] | .946 | .023 |
| Sentence–level comprehension | 743.763 (5) | < .001 | .754 [.690, .820] | .436 | .184 |
| Oral passage reading | 188.439 (14) | < .001 | .181 [.156, .207] | .862 | .066 |
| Numeracy | | | | | |
| Number identification | 468.752 (54) | < .001 | .040 [.037, .044] | .994 | .049 |
| Addition | 283.185 (35) | < .001 | .140 [.123, .158] | .951 | .026 |
| Subtraction | 353.667 (35) | < .001 | .175 [.156, .195] | .921 | .035 |
| Social and emotional learning | | | | | |
| Self-concept[a] | Heywood | | | | |
| Use of social supports[b] | 0.935 (2) | .627 | .000 [.000, .000] | 1.000 | .152 |
| Help seeking behavior[c] | 0.612 (2) | .736 | .000 [.000, .020] | 1.000 | .142 |
| Stress management[d] | | | Heywood | | |
| Empathy[e] | 2443.116 (43) | < .001 | .108 [.105, .112] | .971 | .153 |
| Conflict resolution[f] | Heywood | | | | |

*Note.* Robust RMSEA and robust CFI were reported for listening comprehension, letter/letter sound identification, common word identification, sentence–level comprehension, oral passage reading, addition, subtraction, and use of social supports according to Savalei (2021).
[a]Six items (sel1-sel6) loaded on the latent factor. The tetrachoric correlations between items ranged from.94 to.99. sel6 exhibited negative residual variance.
[b]Four items (rel3, rel4, rel9, and rel13) loaded on the latent factor. The tetrachoric correlation coefficient between rel3 and rel4 was.99.
[c]Four items (rel5, rel6, rel10, and rel14) loaded on the latent factor. The tetrachoric correlation coefficient between rel5 and rel6 was.99.
[d]Four items (rel1, rel2, rel8, and rel12) loaded social support, and three items loaded behavioral regulation (st1-st3). The tetrachoric correlation coefficient between rel1 and rel2 was.99. The tetrachoric correlation coefficients between st1, st2, and st3 ranged from.98 to.99. st3 exhibited negative residual variance.
[e]Five items (rel7, rel11, rel15, e1, and e6) loaded on identifying feelings of others, and six items (e2, e3, e5, e7, e8, and e10) loaded on empathy.
[f]Four items (con1-con4) loaded on social problem solving, and two items (e4 and e9) loaded on interpreting hostility. The tetrachoric correlation coefficients between con1-con4 ranged from.93–.98. con4 exhibited negative residual variance.

**Table 5** ReAL criterion validity spearman correlations by country and literacy domain.

| ReAL subtest | oPt ($n = 65$) | Mozambique ($n = 79$) | Cambodia ($n = 206$) | | | Philippines ($n = 188$) | |
|---|---|---|---|---|---|---|---|
| Letter/Letter Sound Identification | −.042 | .368 | .702* | .694* | .720* | −.058 | −.097 |
| Common Word Identification | .530 | .129 | .782* | .667* | | −.084 | |
| Reading Comprehension | .503 | .059 | .724* | | | .168* | |

*oPt* & Mozambique Criterion = letter sounds, familiar word fluency & reading comprehension
*Cambodia* Criterion = identifying consonants, vowels & consonants subscripts; reading words & sentences; reading comprehension
*Philippines* Criterion = letter name & sound knowledge, familiar word decoding & reading comprehension
*Expressive* vocabulary, listening comprehension & retelling a story lack an adequate criterion match
[*]Indicates significance at *p*-value < .05, only calculated for countries with sample sizes >100

also found in oPt ($r_s = .530$), Niger ($r_s = .457$), and a small positive correlation was found in Mozambique ($r_s = .130$). A small negative correlation was found in the Philippines ($r_s = -.084$).

Reading comprehension was significantly and positively correlated with the criterion measure in both Cambodia (Spearman = .724), Niger ($r_s = .257$), and the Philippines. ($r_s = .168$). The sub-domain was positively correlated in oPt ($r_s = .503$), and a small positive correlation was also seen in Mozambique ($r_s = .060$).

## Numeracy

In terms of the numeracy criterion correlations, a positive correlation was seen between all ReAL sub-domains and corresponding criterion sub-domains, except for one-to-one correspondence which was not included due to lack of variability and lack of a corresponding criterion measure. The strongest positive correlations were seen in Cambodia, with correlations ranging from.700 to.789. The weakest correlation was seen between ReAL subtraction and EGMA subtraction in Niger ($r_s = .077$).

## Social-emotional learning

Criterion correlations for the SEL domain were inconclusive. Correlations were often negative, in an unexpected direction, and were mostly insignificant. The only significant correlations were seen between the ReAL empathy sub-domain and the criterion measure domain of solving conflicts in Cambodia ($r_s = .408$), and the ReAL help seeking behavior and the criterion measure domain of social wellbeing ($r_s = .155$) in Niger.

**Table 6** ReAL criterion validity spearman correlations by country and numeracy domain.

| ReAL subtest | oPt ($n = 65$) | Mozambique ($n = 79$) | Cambodia ($n = 206$) | Philippines ($n = 188$) | |
|---|---|---|---|---|---|
| Number Identification | | | .779* | | |
| Addition | .254 | .195 | .736* | .097 | .180* |
| Subtraction | .390 | .196 | .700* | .163* | .089 |
| Word Problems | .542 | .134 | | .210* | |

*One* to one correspondence was not included due to the lack of variability
*Number* identification was not included due to the lack of corresponding counterpart in previously measured criterion measure except in Cambodia where one existed
*Indicates significance at $p$-value < .05, only calculated for countries with sample sizes > 100
*Philippines* criterion measure included a level 1 and level 2 for addition and subtraction

**Table 7** ReAL criterion validity spearman correlations by country and social emotional learning domain.

| | oPt ($n = 65$) | | | Mozambique ($n = 79$) | | | Philippines ($n = 188$) | Cambodia ($n = 201$) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **ReAL subtest** | **E** | **S** | **RC** | **E** | **S** | **RC** | **SEL** | **PT** | **SC** | **H** |
| Self-concept | | | | | | | .023 | | | |
| Use of social supports | | .019 | | | −.087 | | −.053 | | | .0800 |
| Help seeking behavior | | −.0380 | | | −.056 | | .004 | | | .132 |
| Stress management, social supports | .1900 | | .0500 | .271 | | −.009 | −.107 | | | .066 |
| Stress management, behavioral regulation | .142 | | −.152 | −.048 | | .300 | −.012 | | | |
| Identifying feelings of others | .182 | −.015 | | .077 | −.065 | | −.071 | .039 | | |
| Empathy | .224 | | .060 | −.102 | | .242 | −.083 | .019 | .408* | |

*Indicates significance at $p$-value < .05, only calculated for countries with sample sizes > 100.
*Notes.* E = Emotional Wellbeing; S = Social Wellbeing; RC = Resilience/Coping; SEL = social emotional learning; PT = perspective taking; SC = solving conflicts; H = offering help & asking for help.

## Test–retest reliability

Test–retest correlations were calculated for each country by sub-domain (Tables 8, Table 9, and Table 10). Test–retest reliability varied by country and sub-domain but were overall the strongest in the literacy sub-domains across countries, and between countries were overall the strongest in Cambodia.

### Literacy

In terms of the literacy domain, Cambodia's test–retest correlations ranged from fair to excellent (from .462 to.858). In oPt, all test–retest correlations ranged from fair to excellent except for retelling a story, which had a Spearman correlation coefficient of .383. In Niger, all test–retest correlations were fair except for sentence comprehension, which had a Spearman correlation coefficient of .334. In the Philippines, only sentence comprehension and reading comprehension had fair reliability or higher, and in Mozambique, only retelling a story, letter identification, and common word identification had fair reliability or higher.

### Numeracy

For the numeracy domain, Cambodia's test–retest correlations ranged from good to excellent, except for one-to-one correspondence, which had a Spearman correlation of .293. In oPt, all test–retest correlations ranged from fair to excellent. In Niger, only the number identification sub-domain had fair test–retest reliability ($r_s = .490$). In Mozambique only the number identification and addition sub-domains had fair reliability or higher, and in the Philippines none of the sub-domains had fair or higher reliability.

### Social-emotional learning

For social emotional learning, Cambodia's test–retest correlations ranged from fair to good, except for stress management-behavioral regulation, which had a Spearman correlation of .375. In oPt, all test–retest correlations were mostly fair, except for self-concept ($r_s = -.030$), use of social supports ($r_s = .361$), and stress management-behavioral regulation ($r_s = .162$). In Niger and Mozambique none of the sub-domains had fair or higher reliability. In the Philippines only self-concept ($r_s = .560$) and stress management-social supports ($r_s = .458$) had fair reliability.

## Discussion

The goal of the current study was to assess the psychometric properties of the high access modality of ReAL. Specifically, we appraised the inter-rater reliability, underlying factor structure, item slope and difficulty, criterion validity, and test–retest reliability. Our results show, with only a few notable exceptions, moderate evidence that ReAL is a valid and reliable measure of the literacy and numeracy sub-domains assessed. The evidence for the social-emotional sub-domains is less robust. These results lend themselves to revisions, bringing the literacy and numeracy items in line with the skill levels of 5–14-year-old children in the

**Table 8** ReAL test–retest spearman correlation by country and literacy sub-domain.

| | Philippines | | Cambodia | | oPt | | Mozambique | | Niger | |
|---|---|---|---|---|---|---|---|---|---|---|
| **ReAL literacy sub-Domain** | **Sample** | **Correlation** | **Sample** | **Correlation** | **Sample** | **Correlation** | **Sample** | **Correlation** | **Sample** | **Correlation** |
| Expressive vocabulary | 153 | .399 | 210 | .641 | 168 | .647 | 77 | .090 | 171 | .484 |
| Retelling story | 92 | .328 | 159 | .462 | 146 | .383 | 6 | .791 | 82 | .517 |
| Listening comprehension | 153 | .137 | 210 | .490 | 163 | .536 | 54 | .226 | 154 | .442 |
| Letter identification | 153 | −.020 | 210 | .737 | 168 | .769 | 78 | .472 | 188 | .421 |
| Common word identification | 153 | .333 | 198 | .826 | 156 | .703 | 42 | .599 | 184 | .509 |
| Sentence comprehension | 153 | .481 | 189 | .85 | 158 | .530 | 75 | .284 | 184 | .334 |
| Reading comprehension | 138 | .484 | 126 | .595 | 106 | .554 | 14 | .085 | 85 | .516 |

*Note.* For individuals with partial missingness, missing values were assumed to be incorrect. Individuals missing all information on a given sub-domain were dropped.

**Table 9** ReAL test–retest spearman correlation by country and numeracy domain.

| | Philippines | | Cambodia | | oPt | | Mozambique | | Niger | |
|---|---|---|---|---|---|---|---|---|---|---|
| **ReAL numeracy sub-Domain** | **Sample** | **Correlation** | **Sample** | **Correlation** | **Sample** | **Correlation** | **Sample** | **Correlation** | **Sample** | **Correlation** |
| One to one correspondence | 153 | −.009 | 210 | .293 | 167 | .412 | 79 | N/A[a] | 182 | .013 |
| Number identification | 152 | −.045 | 209 | .797 | 166 | .790 | 80 | .680 | 186 | .490 |
| Addition | 152 | .199 | 194 | .783 | 155 | .412 | 70 | .459 | 185 | .332 |
| Subtraction | 152 | .146 | 196 | .757 | 156 | .703 | 63 | .338 | 185 | .282 |
| Word Problems | 152 | .381 | 192 | .657 | 156 | .581 | 68 | .298 | 172 | .189 |

*Note.* For individuals with partial missingness, missing values were assumed to be incorrect. Individuals missing all information on a given sub-domain were dropped.
[a]No variation was observed for the one-to-one correspondence questions in Mozambique. Therefore, a test–retest reliability correlation could not be calculated

**Table 10** ReAL test–retest reliability spearman correlation by country and social emotional learning domain.

| **ReAL social emotional learning sub-Domain** | Philippines | | Cambodia | | oPt | | Mozambique | | Niger | |
|---|---|---|---|---|---|---|---|---|---|---|
| | **Sample** | **Correlation** | **Sample** | **Correlation** | **Sample** | **Correlation** | **Sample** | **Correlation** | **Sample** | **Correlation** |
| Self-concept | 153 | .560 | 210 | .455 | 169 | −.030 | 80 | .292 | 192 | .247 |
| Use of social supports | 153 | .281 | 210 | .561 | 169 | .361 | 80 | −.131 | 192 | .089 |
| Help seeking behavior | 153 | .280 | 210 | .654 | 169 | .546 | 80 | −.041 | 192 | .133 |
| Stress management, social supports | 153 | .458 | 210 | .531 | 169 | .434 | 80 | .135 | 192 | .072 |
| Stress management, behavioral regulation | 153 | .050 | 210 | .375 | 169 | .162 | 80 | −.041 | 192 | .125 |
| Identifying feelings of others | 151 | .292 | 207 | .497 | 165 | .457 | 79 | −.145 | 174 | .217 |
| Empathy | 137 | .140 | 206 | .618 | 165 | .598 | 79 | −.003 | 174 | .131 |

*Note.* For individuals with partial missingness, missing values were assumed to be incorrect. Individuals missing all information on a given sub-domain were dropped.

countries that participated in the study through further testing and contributions from country partners.

## Implications of different results by academic versus non-academic domains

The results for literacy and numeracy demonstrate moderate evidence for valid and reliable measures of these skills. Except for expressive vocabulary, retelling a story, sentence-level comprehension, and one-to-one correspondence, we find reasonable inter-rater reliability, construct validity, criterion validity, and test–retest reliability. Expressive vocabulary reliability and validity results were excellent for inter-rater reliability, but in terms of test–retest, evidence varied substantially by country, with inadequate results in Mozambique and good to fair results in the remaining countries. Similarly, retelling a story had satisfactory inter-rater reliability correlations, but evidence varied substantially by country in terms of test–retest, with fair results in Cambodia, and Niger, and unsatisfactory results in the remaining countries. Sentence comprehension fared worse, with only inter-rater reliability results found to be consistently satisfactory, as unsatisfactory fit statistics from the CFA were found, and only fair to good test–retest results were found in oPt, the Philippines, and Cambodia. One-to-one correspondence had the worst test–retest results, with only oPt having fair correlations. As one-to-one correspondence is one of the easiest subtests in ReAL, minor variations in response to the three items may account for the low correlation coefficients due to the restriction of range. Several unobserved factors could have accounted for the low test–retest reliability. The reliability of ReAL over time depends on whether the responses of the children were consistent over time. In other words, the test–retest reliability estimate captures the extent to which the results can be reproduced under the same assessment conditions. Although the administration guidance of ReAL is standardized and, as evidenced by the inter-rater reliability, items can be applied accurately by trained data collectors, the guidance does not stipulate the exact assessment conditions. In some cases, for example, when the cell phone connectivity was unstable, the data collectors retried contacting the family over the course of the day sometimes leading to assessments during a different time of day, which could affect the child's performance. Further, sometimes different data collectors were present across occasions. The IIFs and TIFs suggest that making items more difficult in all sub-domains while retaining the assessment approach could precisely assess a wide range of abilities.

The results for social and emotional skills are more complex. Despite many sub-domains performing well in face-to-face assessments (D'Sa & Krupar, 2021; Krupar & D'Sa, 2024), there is inconclusive or poor evidence supporting the current form of social and emotional sub-domains in ReAL. Items assessing numeracy and literacy skills are arguably more homogenous in their presentation and (cognitive) demands to solve the respective problem than social and emotional skills due to their straightforward and standardized nature, allowing for consistent measurement and comparability across different countries and cultures (Kaffenberger & Pritchett, 2020). Because the acquisition of social and emotional skills is context- and age-dependent (McCoy, Cuartas, Waldman, & Fink, 2019), the assessment requires significant adaptation to be accurately assessed in different cultural settings (Schonert-Reichl et al., 2015). The subjective nature of these assessments introduces additional complexity, emphasizing the need for comprehensive enumerator training (Humphrey, 2013). Enumerators in Sudan shared that although the questions asked in the social and emotional skills domain were child-friendly, they were quite sensitive for children who had experienced conflict in their lifetime. Fear or a desire to appear "normal" might have led the children to downplay their true feelings and experiences, especially to enumerators who were unfamiliar to the children. In Niger, enumerators shared that social and emotional skills are not integrated into the education curriculum, and children are not accustomed to being questioned about their social-emotional wellbeing. As a result, many children remained silent during questioning for this section, likely due to a lack of understanding or not knowing how to respond. Enumerators shared a similar sentiment in Cambodia and spoke to how SEL is only integrated into the curriculum in specific UNICEF-funded schools, and most schools do not teach any SEL curriculum. Colleagues in the Philippines shared that the SEL curriculum has not been mainstreamed into the education framework, that teachers are not trained in SEL, and that there are uncoordinated efforts among sectors and agencies. Given that ReAL is designed to access hard-to-reach children, these considerations will be central for future iterations and revisions of the tool.

Administering these assessments remotely further complicates the process, especially for SEL. Remote assessments lack the direct, in-person interaction crucial for accurately gauging social and emotional competencies. Enumerators may struggle to interpret non-verbal cues or create a supportive environment conducive to honest self-expression when not physically present. This challenge is compounded by the lack of substantial evidence on the effectiveness and reliability of remote SEL assessments, as most existing studies and tools are designed for in-person administration (Domitrovich et al., 2017). This shift highlights the urgent need for research to develop and validate reliable methods for remotely assessing SEL skills. However, the complications due to the remote administration influence all domains of ReAL. For example, enumerators in Mozambique shared that although children often recognize lowercase and uppercase letters when they are handwritten, they may not be able to recognize the typed letter on a screen. Together, the results showed that some tasks yielded psychometric properties that did not meet conventional cutoffs or benchmarks. First, despite the comprehensive and rigorous enumerator training, we cannot rule out implementation quality differences related to supervision and fidelity monitoring across sites. Second, the low inter-rater reliability for SEL tasks in some countries may point to linguistic and cultural factors such that the items do not translate equivalently across countries, and children's familiarity with the specific type of question may vary. Third, while the age range was pre-determined (5–14 years), the country samples differ in the de facto demographic characteristics, including age ranges and other characteristics such as exposure to formal education and socioeconomic diversity. Fourth, connectivity issues and disruptions may have affected some sites more than others, leading to potentially diverging results across contexts (e.g., if one of two enumerators experiences occasional weak connectivity). Fifth, the enumerators' background and experience may account for differences in psychometric properties across countries. For instance, previous experience administering standardized assessments, experience working with marginalized populations, and familiarity with the age group could inadvertently affect assessment quality.

## Implications of limited variability across target age range

Appraising the IIFs and TIFs in our study shows that several items and sub-domains mainly provide information for children performing below the mean. This phenomenon is most apparent among younger children due to their varying stages of cognitive and emotional development, which influence their learning abilities and skill acquisition. Younger children are earlier in their development, which can lead to significant differences in their literacy, numeracy, and social and emotional skills. The early years are crucial for language development, and children's exposure to language-rich environments significantly impacts their literacy skills (Shonkoff & Phillips, 2000). Similarly, early experiences with numbers and spatial reasoning, which vary widely depending on the child's home environment and parental engagement, affect numeracy skills (Ginsburg et al., 2008).

Social and emotional skills are context-dependent and develop through interactions with caregivers and peers, leading to variability in children's social competencies and emotion regulation skills (Denham et al., 2012). Younger children are more susceptible to adverse experiences like poverty and stress, impacting their development in these domains (Yoshikawa et al., 2012). As children grow older, more structured education helps mitigate early disparities. Older children have had more time to develop compensatory skills and strategies, reducing variability and disparities seen in younger age groups (Entwisle & Alexander, 1993). In contrast, foundational skills in literacy and numeracy begin with lower-order skills essential for acquiring higher-order competencies. In literacy, this starts with phonemic awareness and phonics (National Institute of Child Health and Human Development, 2000). As children progress, they develop higher-order literacy skills, such as fluency, vocabulary, and comprehension, enabling them to read with speed, accuracy, and expression, and to understand and interpret complex texts (Snow, 2002). In numeracy, lower-order skills begin with number sense and basic arithmetic (Berch, 2005). We propose retaining the wide age range to increase applicability in LMIC contexts, as older children may still be mastering foundational skills due to lack of access or disruptions to school. We also suggest revising the measure to include more difficult items for higher-performing children while dropping the sub-domains retelling a story, sentence-level comprehension, and one-to-one correspondence due to their poor performance across age groups. We also propose exploring the feasibility of making ReAL an adaptive assessment.

## Implications, limitations, and future directions

Through this study, we demonstrate that a remote assessment of learning can be a feasible, valid, and reliable measure of foundational academic (i.e., literacy and numeracy) skills in LMICs. While this evidence is promising, there are critical preconditions that must be met when conducting a remote, phone-based assessment: a cellular network infrastructure and connectivity and ownership of or access to a phone. These conditions may not always be present in many low-resource contexts. For this reason, it is important to understand the context in which the assessment will take place to evaluate whether such an assessment is appropriate. Thus, we advocate for assessments like ReAL to complement existing approaches used by education systems to assess foundational skills, rather than serve as replacements. Based on our findings, we recommend ReAL for screening applications in combination with other data sources, and for monitoring the implementation quality of educational programs in hard-to-reach contexts. Due to the mixed results concerning some psychometric properties, we caution against using ReAL for high-stakes individual decisions given low test–retest reliability in some contexts, and contexts in which adequate training and supervision cannot be provided.

There are several limitations of this study that are worth noting. In Mozambique, due to low performance of some enumerators, half of the enumerators were replaced between test–retest observations. As such, the low test–retest correlations in Mozambique may be partly due to enumerator differences, not time differences. In Sudan, data collection was paused due to the civil war and therefore only inter-rater reliability, and CFA data could be collected. There was also the possibility of greater-than-average parental involvement in the assessment process. For example, in Cambodia enumerators shared that although the project team clearly articulated the purpose of the ReAL validation, some caregivers attempted to whisper answers and correct the children's mistakes during data collection and instances of children being blamed were also overheard during the on-call assessments. However, most countries reported none to minimal parental involvement (i.e., in Niger enumerators estimated approximately 5% of parents attempted to influence the results). We cannot confirm the level of involvement and propose measuring this in future studies.

Given the promise of remote, phone-based assessments as one option to assess foundational skills of hard-to-reach children in LMIC contexts, we advocate for further research that builds upon this validation study. Specifically, we propose developing and testing adaptive versions that could more efficiently assess skills across the wide age range we target here and those that look at variation by sex and gender. Additionally, building evidence of cost effectiveness in comparison with other assessment types will be a critical consideration for any education system weighing a phone-based assessment like ReAL with one administered face-to-face. Moreover, the lack of concurrent in-person assessments is a fundamental methodological constraint. Data from such assessments would provide additional evidence about ReAL's convergent/discriminant validity. Finally, we seek to re-develop and test the ReAL SEL sub-domains in future studies. Additional revisions of ReAL are underway. Interested readers and prospective users may refer to the following website for guidance documents and more information (Save the Children International, 2021).

## Conclusion

The COVID-19 pandemic forced the global community to face a persistent gap that has been present in our foundational skills assessment toolbox and thus continued to limit our ability to deliver on the promise of SDG 4: our inability to assess hard-to-reach children when face-to-face access was not possible. The pandemic catalyzed the development and testing of phone-based, remote assessments in LMICs, providing us with some promising options. While the evidence remains thin, we are beginning to coalesce some promising examples. These phone-based assessments of foundational learning skills have the potential to expand education systems' knowledge of the skill levels of even the hardest-to-reach children, providing critical information for education decision makers. This study contributes to the evidence base on valid and reliable measures of foundational academic skills for children 5–14 years old and provides directions for future development of valid and reliable measures of foundational SEL skills.

## Supplementary material

Supplementary material is available at *Child Development* online.

## Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

## Author contributions

Elizabeth Hentschel (Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing—review & editing [equal], Writing—original draft [lead]), Sascha Hein (Data curation, Formal analysis, Funding acquisition, Methodology, Supervision, Writing—original draft, Writing—review & editing [equal]), Nan Li (Formal analysis, Validation, Visualization, Writing—original draft, Writing—review & editing [equal]), Clay Westrope (Conceptualization, Data curation, Funding acquisition, Investigation [lead], Formal analysis, Methodology, Writing—original draft, Writing—review & editing [supporting], Resources, Validation [equal]), Julia Taladay (Data curation [equal], Formal analysis, Funding acquisition, Investigation, Methodology, Writing—original draft, Writing—review & editing [supporting]), Gillian Valentine (Funding acquisition, Methodology, Project administration [equal]), James Leckman (Writing—review & editing [equal]), Amira Abdurahman (Project administration [lead], Writing—review & editing [supporting]), Fatime Bachir (Project administration [lead], Writing—review & editing [supporting]), Farah Sadi Darwazeh (Project administration [lead], Writing—review & editing [supporting]), Xerxes M De Castro (Project administration [lead], Writing—review & editing [supporting]), Doa Hamdan (Project administration [lead], Writing—review & editing [supporting]), Sithon Khun (Project administration [lead], Writing—review & editing [supporting]), Sakem Kong (Project administration [lead], Writing—review & editing [supporting]), Harouna Mounkaila (Project administration [lead], Writing—review & editing [supporting]), Janet Mugo (Project administration [equal], Writing—review & editing [supporting]), Mohamed Sagayar (Project administration [lead], Writing—review & editing [supporting]), Ali Nashat Shaar (Project administration [lead], Writing—review & editing [supporting]), Borin Srey (Project administration [lead], Writing—review & editing [supporting]), Adriano S Uaciquete (Project administration [lead], Writing—review & editing [supporting]), Nelson Zavale (Project administration [lead], Writing—review & editing [supporting]), and Liliana Angelica Ponguta (Conceptualization, Formal analysis, Funding acquisition, Investigation, Writing—review & editing [supporting])

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## Conflicts of interest

The authors declare no conflict of interest

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